Time Series Forecasting: An Exploratory Study Using S&P 500 ETF Data

Gina Occhipinti
Woods College, Master's of Science in Applied Economics, Boston College

Abstract

Background

Forecasting is a valuable tool in creating intelligent estimates about the future. This is especially pertinent in finance, a key industry of the United States economy. Understanding the past, present, and future movements of financial instruments like stocks, bonds, and exchange traded funds (ETFs) is extremely valuable for maximizing investor profits. It is challenging, however, because instrument price movements are notoriously volatile. This study uses the popular ETF SPY-USA, a market cap-weighted index of US large- and mid-cap stocks, to explore how forecasting methods can predict future closing prices.

Methodology

Monthly price data spanning 10 years (2015 to 2025) for the SPY-USA was anaylzed from FactSet. The data was transformed for modeling and visualized with seasonal-trend using LOESS (STL) decomposition to reveal underlying patterns not seen in the raw data. The following models were then tested to forecast the SPY-USA ETF price: exponential smoothing (ETS), Naïve, Seasonal Naïve, and Ensemble.

Results

The additive and multiplicative ETS models (both Holt linear and Holt–Winters) and the basic naïve model delivered the smallest forecast errors (MAE and RMSE) and residuals consistent with assumptions. The seasonal naïve approach under-predicted the strong upward trend and exhibited autocorrelated residuals, making it the least reliable. The ensemble model offered modest gains over its weak seasonal component but exhibited model bias.

Introduction & Significance

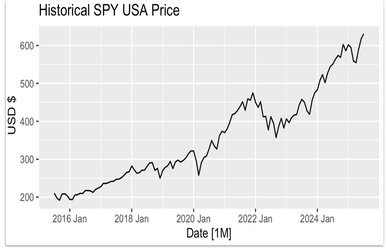
There are some theories on how to predict market performance. Yates (2024) discusses momentum theory. Rooted in behavioral finance, it assumes that market movements will continue in the same direction, but only in the short-term. Over longer periods, this effects reverses. Pande and Kumar (2025) acknowledged traditional methods to forecast stock prices and indices such as stochastic time series models (e.g., ARIMA, jump diffusion) but also modern machine learning methods (e.g., FFN, LSTM). The study found that LSTM models performed best for forecasting over almost all indices over all time horizons except for the S&P 500 and Nikkei 225 10 days ahead, where ARFIMA performed best.

Shih et al. (2024) wrote a journal article comparing deep learning models to more traditional models, such as Fama-French 3-factor models. This study found similar success in LSTM models and used them in combinations with the Fama-French models. Ngwaba (2025) takes a unique approach, being one of the first to focus on forecasting applications specifically for covered call ETFs and also analyzes machine and deep learning models. Here, RNNs outperformed other models with the lowest MAE.

Limitations in existing literature recognize that deep learning models require significant computation demands, extensive data needs, and fail to capture many key factors that influence stock and ETF prices, like macroeconomic and geopolitical events.

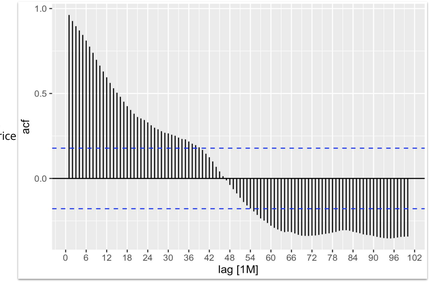
Methods

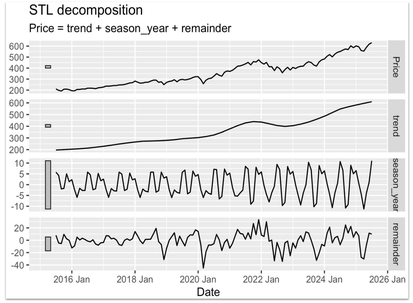
During data preparation, I reduced the variables to only the temporal variable (month and year) and the price, referring to the closing price or the final traded price when the market closed that day. Additional cleaning involved mutating the temporal variable to a monthly format and identifying any missing values. Preparation also involved visualizing the data to understand any patterns like trends or seasonality through STL decomposition. This provided a deeper understanding of the data to best understand how the models performed and when interpreting the results. I also tested using inflation adjusted price values (using Core CPI), however, it did not impact any patterns in the data. The data was split 80/20 into training and testing data, respectively.



Historical SPY USA Price has an upward trend, no seasonality and some irregular variation.

Data has strong autocorrelations, typical of stock price data.



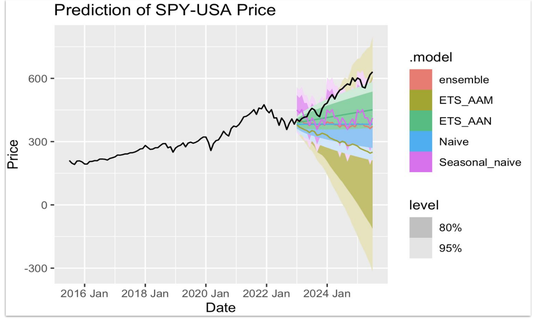


Shows a smooth upward trend. Seasonality with increasing magnitude (multiplicative effect).

Results

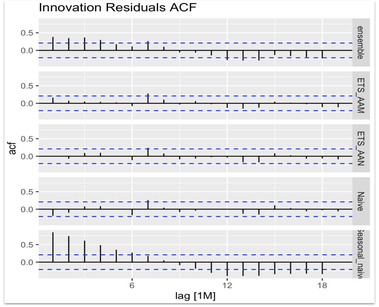
Several models were used for comparison to determine which is the best for predicting the SPY: Exponential Smoothing State Space Model (ETS) with automated parameter selection (with extensions in Holt Linear Trend and Holt-Winters), a naïve model, seasonal naïve model, and an ensemble model that combined forecasts using averaging.

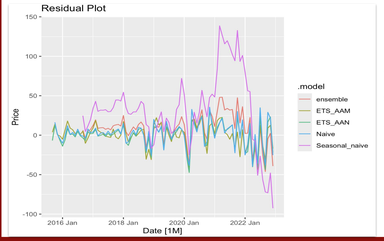
Overall, the ETS and naive models performed best.
ETS_AAM had the best (smallest) error. On average, the ETS_AAM model misses the true value of the SPY price by about 4%. The ensemble model ranked closed to last in the plots, residual checks, and accuracy metrics, likely being affected by the poor results of the seasonal_naive model being part of the ensemble, to capture the average. Regarding MAPE, all models generally performed well.



ETS and Naïve models performed best while Seasonal Naïve performed worst.

Most residuals stayed within acceptable bounds, behaving like white noise, except for Seasonal_Naive and Ensemble.





Few residuals hovered around 0, especiallly Seasonal_Naive , indicating there's something systematically biasing the residuals, violating a key assumption.

Discussion

The forecasting process involves building from numerous options models that fit the data and capture events in order to estimate what will happen in the future with accuracy. It's important to consider the data structure and behavior to pick the best models. In comparing different models, it's also key to benchmark against a baseline, such as with naïve models or seasonal naïve models, when appropriate.

This study considers a popular financial investment, the SPY-USA ETF, that tracks the S&P 500 and the different models that can effectively forecast performance. This study focused on low computational forecasting methods for tech savvy investors. It also explore an ensemble model, something mentioned as an area of further exploration by other authors in the field.

Some limitations and challenges in this study include not considering more advanced machine or deep learning models that other studies analyzed. While it does explore ensemble methods that average results from many models, it doesn't include hybrid models that may have produced better results, like ETS_AAN and LSTM. Another limitation is it doesn't consider other factors that could influence ETF performance that are challenging to model, such as involvement in war, tariffs, and other geopolitical factors.

Investors should consider simple and moderately complex models like naïve and exponential smoothing as information to incorporate into investment decisions, like when to invest in the S&P 500. However, investors should not rely solely on these forecasts and incorporate judgmental forecasting and domain knowledge to improve results.

Conclusion

This study compares forecasting methods on 10 years of monthly SPY-USA data. Overall, the additive and multiplicative ETS models and the basic naïve model delivered the smallest forecast errors (MAE and RMSE) and residuals consistent with assumptions. Often, simple models suffice when it comes to financial forecasting. Limitations of this work include its focus on univariate historical prices and omission of exogenous factors as well as more advanced machine and deeplearning approaches.

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